MICROSCALE OCEAN BIOPHYSICS 8.0

Workshop Program: Thursday May 28, 2026 - Saturday, May 30, 2026

**Several workshops will require a personal laptop with a recent release of Matlab (post 2020) and the following Toolboxes: Image Analysis Toolbox and Curve Fitting Toolbox. It would be helpful to install them before arriving. We will also be following up with software to download for different days of the workshops. Material for the workshops will be provided in this repository on GitHub.

Details of the workshops are included in the following pages:
SNO: The methods and theory of marine particle aggregation, sedimentation, and rheology
PAM: Photosynthetic parameters from chlorophyll autofluorescence
MPP: Changes in algal motility caused by light exposure
MBF: Sampling and observing benthic microorganisms forming biofilms

Participants will be divided into 4 teams (A,B,C,D) and each team will partake in a different workshop each day with the following schedule:

TEAM	DAY 1 (28 MAY Thursday)		DAY 2 (29 MAY)	DAY 3 (30 MAY)	
A	SNO1	SNO1	MBF1	PAM3	MPP4
B	SNO1	SNO1	MBF1	MPP3	PAM4
C	PAM1	MPP2	MBF1	SNO2	SNO2
D	MPP1	PAM2	MBF1	SNO2	SNO2

The methods and theory of marine particle aggregation, sedimentation, and rheology (SNO)

Leads:
Òscar Guadayol, Mediterranean Institute for Advanced Studies, IMEDEA, Spain
oguadayol@imedea.uib-csic.es
Matthew Rau, George Washington University, USA
matthew.rau@gwu.edu
Bryce Inman, Stanford University
binman@stanford.edu

Overview: This workshop will provide a hands-on introduction to the study of marine particulate matter with a focus on key physical properties related to understanding their transformation and fate in the oceanic water column. Experimental methods to create marine snow particles in a laboratory setting, understand particle aggregation and disaggregation rates, compare particle cohesiveness, measure aggregate sedimentation velocities, and quantify particle microrheological properties will be explained and demonstrated through hands-on participant experiments. The background theory inherent to these methods will also be explained, and other community-developed tools and methodologies to study related marine snow processes will also be introduced. Participants should expect to come away from this workshop with a better understanding of experimental tools to study marine snow physical properties, their limitations, and how they can be applied to advance our knowledge of the ocean biological carbon pump.

Schedule:

08:30-09:15: Session overview

09:15-10:00: Seawater sample collection

10:00-10:30: Break

10:30-12:00: Rotate stations

- Station A: Aggregation / Disaggregation
- Station B: Particle Sedimentation
- Station C: Marine Snow Microrheology

12:00-14:30: Lunch break

14:30-16:00: Rotate stations

16:00-16:30: Break

16:00-18:00: Rotate stations

Photosynthetic parameters from chlorophyll autofluorescence (PAM)

Leads:
Antoine Allard, Université de Bordeaux, France
antoine.allard@u-bordeaux.fr
Ludovic Jami
ljami@imedea.uib-csic.es

Overview: Here we provide an introduction to the physical and biological basis of chlorophyll fluorescence and its use as a proxy for photosynthetic activity in microalgae. The session introduces the principles of Pulse Amplitude Modulation (PAM) fluorometry, including the role of measuring, actinic, and saturating light, and the definition of photosynthetic parameters. Participants will learn how to acquire, visualize, and interpret PAM fluorescence signals in microalgal samples, and how these measurements relate to photosynthetic efficiency. It includes:

1) basis of chlorophyll fluorescence and its use as a proxy for photosynthetic activity;
2) the principles of Pulse Amplitude Modulation (PAM) fluorometry and the meaning of commonly used photosynthetic parameters (F0, Fm, Fv/Fm);
3) the acquisition and interpretation of PAM fluorescence signals in microalgal samples.

Schedule:

08:30-09:15 (14:30-15:15): Welcome and introduction to the session

09:15-10:00 (15:15-16:00): Practical: Autofluorescence parameters of dark-adapted cells, then at various actinic lights.

10:00-10:30 (16:00-16:30): Break

10:30-12:00: (16:30-18:00): Continue Practical

Light-Driven Motility in Microalgae (MPP)

Leads:

Ahmad Badr, Université de Bordeaux, France
ahmad.badr@u-bordeaux.fr
Sujeet Kumar Choudhary, Spain
sujeet@imedea.uib-csic.es

Overview: This workshop investigates the motile behaviour of unicellular microalgae in response to light stimuli, with a focus on two distinct photomotility phenomena: **phototaxis** and **photokinesis**. Phototaxis refers to directional locomotion oriented along a light gradient - either towards the source (positive phototaxis) or away from it (negative phototaxis). Photokinesis, by contrast, is a light-induced change in swimming speed that is non-directional. Participants will gain hands-on experience in microscopy-based cell tracking, quantitative motility analysis, and the design of controlled light-stimulation assays.

For each experimental session, participants will be divided into two subgroups, each assigned to one of the two experimental modules: 1. Phototaxis, 2. Photokinesis. Both modules largely use similar experimental methods, except for the wavelength of the light stimulus.

In this activity,

- We will use a simple microscope equipped with a camera to record the motility of microalgae confined within a quasi-2D chamber.
- We will first observe cell motility under far-red background illumination and then compare it with conditions in which the cells are additionally illuminated laterally using a UV LED (for photokinesis)/green LED (for phototaxis).
- We will introduce UV/green light at varying intensities following dark adaptation between two consecutive intensities.
- Finally, we will compare photokinetic/phototactic behavior at different light intensities and motility during the periods between light stimulations.

Together, these investigations deepen our understanding of how light shapes navigation, metabolic activity, and communication in the microbial world.

Schedule:

08:30-09:15 (14:30-15:15): Welcome and introduction to the session

09:15-10:00 (15:15-16:00): Practical: Photomotility. Cell phototaxis towards/away from light; photokinesis as a consequence of UV exposure.

10:00-10:30 (16:00-16:30): Break

10:30-12:00: (16:30-18:00): Continue Practical

Sampling and observing benthic microorganisms forming biofilms (MBF)

Leads:
Elisa Berdalet, Institute of Marine Sciences (ICM-CSIC), Barcelona
berdalet@icm.csic.es
Salima Rafai, University of Grenoble-Alpes
salima.rafai@univ-grenoble-alpes.fr

Overview: Here we will provide an introduction to:

1. Benthic dinoflagellates (theory):
   - Theoretical introduction: cells both attached to substrates and swimming; general ecology (presentation, bibliography, some movies)
   - Microscope observations of some cultures
   - Introduction to sampling benthic dinoflagellates procedure
2. Field sampling of benthic dinoflagellates and observation (if successful).
3. Quantitative imaging of plankton using Planktoscope: a low cost tool that combines microscopy and fluidic for imaging plankton

Planktoscope resources online: https://dx.doi.org/10.17504/protocols.io.bp2l6bq3zgqe/v5

Schedule: To be adapted depending on logistics

08:30-10:00: Teams A and B: Benthic dinoflagellates (Theory and sampling)
Teams C and D: Field sampling plankton net

10:00-10:30: Break

10:30-12:00: Teams A and B: Benthic field sampling and observations
Teams C and D: Imaging using Planktoscope

12:00-14:30: Lunch break

14:30-16:00: Teams A and B: Field sampling plankton net
Teams C and D: Benthic dinoflagellates (Theory and sampling)

16:00-16:30: Break

16:00-18:00: Teams A and B: Imaging using Planktoscope
Teams C and D: Benthic field sampling and observations